

Potential Outcomes and Experimental Research Design

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bookdown::render_book()

Overview

Topics:

1. Potential Outcomes and Calculating Treatment Effects
2. Experimental Research Design
3. Randomized Inference

Chapter 1

Potential Outcomes and Calculating Treatment Effects

We will use a hypothetical example to show average treatment effects (ATE), average treatment effects on the treated (ATE_T), and the average treatment effects on the untreated (ATE_U). We have potential outcomes of wages for workers for both Y^1 and Y^0 . In real life we only realize one outcome, but we will do this for pedagogical purposes.

```
clear
set more off
cd "/Users/Sam/Desktop/Econ 672/Data"
use potential_outcomes_example, clear
```

The ATE_U is -1.2.

1.1 Calculate the ATE

First we will calculate the average treatment effects (ATE). We can calculate individual treatment effects under this hypothetical example and find the average treatment effects. What we calculate is $ATE = E[Y^1] - E[Y^0]$

We will find the mean for Y^1 and the mean for Y^0 , and then subtract μ_1 from μ_0 .

	WorkerID	Y1	Y0	Treatment	Delta
1	1	817	800	1	17
2	2	945	759	1	186
3	3	823	900	0	-77
4	4	545	600	1	-55
5	5	911	789	1	122
6	6	756	800	0	-44
7	7	600	450	0	150
8	8	713	555	1	158
9	9	659	600	0	59
10	10	600	745	0	-145
11	11	769	701	1	68
12	12	800	734	1	66
13	13	900	910	0	-10
14	14	945	789	1	156
15	15	876	800	1	76
16	16	680	700	0	-20
17	17	815	800	0	15
18	18	675	450	1	225
19	19	800	815	0	-15
20	20	775	700	0	75

Figure 1.1: Potential Outcomes


```
egen mean_y1 = mean(Y1)
egen mean_y0 = mean(Y0)
gen ate = mean_y1 - mean_y0

list ate mean_y1 mean_y0 in 1/1
```

	ate	mean_y1	mean_y0
1.	50.35	770.2	719.85

We estimate the average treatment effect to be 50.35.

1.2 Calculate the ATET

Next, we can calculate the average treatment effect on the treated (*ATET*). We will subset only for those who actually received treatment. What we are calculating is $ATET = E[Y^1|D = 1] - E[Y^0|D = 1]$.

```
egen mean_y1_d1 = mean(Y1) if Treatment == 1
egen mean_y0_d1 = mean(Y0) if Treatment == 1
gen att = mean_y1_d1 - mean_y0_d1

list att mean_y1_d1 mean_y0_d1 in 1/1
```

(10 missing values generated)

(10 missing values generated)

(10 missing values generated)

	att	mean_y1_d1	mean_y0_d1
1.	101.9	799.6	697.7

Our *ATET* is 101.9.

1.3 Calculate the ATEU

Finally, we will estimate the average treatment effect on the untreated (*ATEU*). We will subset for those who did not receive treatment. We are calculating $ATEU = E[Y^1|D = 0] - E[Y^0|D = 0]$. In real life, we do not know $E[Y^1|D = 0]$

```
egen mean_y1_d0 = mean(Y1) if Treatment == 0
egen mean_y0_d0 = mean(Y0) if Treatment == 0
gen atu = mean_y1_d0 - mean_y0_d0

list atu mean_y1_d0 mean_y0_d0 in 3
```

(10 missing values generated)

(10 missing values generated)

(10 missing values generated)

```

+-----+
|  atu   mea~1_d0   mea~0_d0 |
+-----+
3. | -1.2       740.8       742 |
+-----+
```

The *ATEU* is -1.6

1.4 Calculate the SDO

We need to compare $E[Y^1|D = 1]$ and $E[Y^0|D = 0]$ to calculate the simple difference in outcomes. Since we have already calculated $E[Y^1|D = 1]$ and $E[Y^0|D = 0]$, we will just fill in the blanks with some Stata data management.

First, we will set up a duplicate column for $E[Y^1|D = 1]$. Since we have a value for $E[Y^1|D = 1]$, we can push the value forward with `[_n-1]` or the prior value of the observation.

```
gen mean_y1_d1_a = mean_y1_d1
replace mean_y1_d1_a = mean_y1_d1_a[_n-1] if missing(mean_y1_d1_a)
```

Next, we will set up a duplicate column for $E[Y^0|D = 0]$. Since we do not have a value for $E[Y^0|D = 0]$ in the first observation, we will need to add another line of code. We have $E[Y^0|D = 0]$ in the last value, so we can use `[_N]` to get that observation and copy it to the first observation. Next, we will push the values forward similar to the prior example.

	mean_y1_d1	mean_y0_d1	att	mean_y1_d0	mean_y0_d0	atu	mean_y1_d1_a
1	799.6	697.7	101.9	.	.	.	799.6
2	799.6	697.7	101.9	.	.	.	799.6
3	.	.	.	740.8	742	-1.2	799.6
4	799.6	697.7	101.9	.	.	.	799.6
5	799.6	697.7	101.9	.	.	.	799.6
6	.	.	.	740.8	742	-1.2	799.6
7	.	.	.	740.8	742	-1.2	799.6
8	799.6	697.7	101.9	.	.	.	799.6
9	.	.	.	740.8	742	-1.2	799.6
10	.	.	.	740.8	742	-1.2	799.6
11	799.6	697.7	101.9	.	.	.	799.6
12	799.6	697.7	101.9	.	.	.	799.6
13	.	.	.	740.8	742	-1.2	799.6
14	799.6	697.7	101.9	.	.	.	799.6
15	799.6	697.7	101.9	.	.	.	799.6
16	.	.	.	740.8	742	-1.2	799.6
17	.	.	.	740.8	742	-1.2	799.6
18	799.6	697.7	101.9	.	.	.	799.6
19	.	.	.	740.8	742	-1.2	799.6
20	.	.	.	740.8	742	-1.2	799.6

Figure 1.2: Replace with lags

```

gen mean_y0_d0_a = mean_y0_d0

replace mean_y0_d0_a = mean_y0_d0_a[_N] if missing(mean_y0_d0_a[1])
replace mean_y0_d0_a = mean_y0_d0_a[_n-1] if missing(mean_y0_d0_a)

```

Finally, we can calculate the simple difference in outcomes (*SDO*).

```

gen sdo = .
replace sdo = mean_y1_d1_a - mean_y0_d0_a
list ate sdo if _n == 1

```

(20 missing values generated)

(20 real changes made)

```

+-----+
|   ate   sdo |
+-----+
1. | 50.35  57.6 |
+-----+

```

The *SDO* is upwards biased by 6.25 dollars in this hypothetical example.

	mean_y1_d0	mean_y0_d0	atu	mean_y1_d1_a	mean_y0_d0_a
1	.	.	.	799.6	742
2	.	.	.	799.6	.
3	740.8	742	-1.2	799.6	742
4	.	.	.	799.6	.
5	.	.	.	799.6	.
6	740.8	742	-1.2	799.6	742
7	740.8	742	-1.2	799.6	742
8	.	.	.	799.6	.
9	740.8	742	-1.2	799.6	742
10	740.8	742	-1.2	799.6	742
11	.	.	.	799.6	.
12	.	.	.	799.6	.
13	740.8	742	-1.2	799.6	742
14	.	.	.	799.6	.
15	.	.	.	799.6	.
16	740.8	742	-1.2	799.6	742
17	740.8	742	-1.2	799.6	742
18	.	.	.	799.6	.
19	740.8	742	-1.2	799.6	742
20	740.8	742	-1.2	799.6	742

Figure 1.3: Copy the Last observation to the first observation

Chapter 2

Experimental Research Design

We can estimate the *ATE* using experimental research designs. If the experiments were well-designed and well-implemented experiments, then we can use the *ttest* command to calculate the simple difference in outcomes or *SDO*

2.1 Job Corps (2001) Evaluation

The Department of Labor wanted an evaluation of the Job Corps program. The Job Corps program serves disadvantaged youth to improve employment outcomes. Individuals were randomly assigned to Job Corps or out of Job Corps due to limited slots.

2.1.1 Inspect the Data

Let's inspect our outcome variable and treatment variable

```
cd "/Users/Sam/Desktop/Econ 672/Data"
use JC, clear

describe earny4
describe assignment

sum earny4
tab assignment
```

/Users/Sam/Desktop/Econ 672/Data

(Written by R.)

variable name	storage type	display format	value label	variable label
earn4	double	%9.0g		Earnings 4 years after assignment

variable name	storage type	display format	value label	variable label
assignment	double	%34.0g	assignment1	Random assignment to Job Corps

Earnings 4 years after assignment				
Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	9,240
25%	43.50582	0	Sum of Wgt.	9,240
50%	181.7103		Mean	207.6163
		Largest	Std. Dev.	194.5507
75%	307.3206	1821.865		
90%	442.1254	1859.233	Variance	37849.99
95%	544.7895	2357.738	Skewness	1.890325
99%	835.3426	2409.909	Kurtosis	12.32882

Random assignment to Job Corps	Freq.	Percent	Cum.
Randomly assigned out of Job Corps	3,663	39.64	39.64
Randomly assigned to Job Corps	5,577	60.36	100.00
Total	9,240	100.00	

Our outcome variable shows that the median earnings is \$181.71, while the mean is \$207.62.

Our treatment variable has 5,577 assigned to treatment and 3,663 assigned to the control group.

2.1.2 Estimate the impact

We can estimate the simple difference in outcomes with the *ttest* command.

```
ttest earny4, by(assignment)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Randomly	3,663	197.9258	3.072182	185.9368	191.9025	203.9492
Randomly	5,577	213.981	2.675004	199.7675	208.7369	219.225
combined	9,240	207.6163	2.023936	194.5507	203.6489	211.5836
diff		-16.05513	4.134466		-24.15959	-7.950661
diff = mean(Randomly) - mean(Randomly)				t = -3.8832		
Ho: diff = 0				degrees of freedom = 9238		
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 0.0001		Pr(T > t) = 0.0001		Pr(T > t) = 0.9999		

Next, let's run an OLS regression with and without robust standard errors

```
reg earny4 i.assignment
reg earny4 i.assignment, robust
```

Source	SS	df	MS	Number of obs	=	9,240
Model	569892.708	1	569892.708	F(1, 9238)	=	15.08
Residual	349126136	9,238	37792.3941	Prob > F	=	0.0001
				R-squared	=	0.0016
				Adj R-squared	=	0.0015
Total	349696029	9,239	37849.9869	Root MSE	=	194.4

	earn4	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
	assignment						
Randomly assigned to Job Corps		16.05513	4.134466	3.88	0.000	7.950661	24.15959
_cons		197.9258	3.212061	61.62	0.000	191.6295	204.2222

Linear regression

Number of obs	=	9,240
F(1, 9238)	=	15.53
Prob > F	=	0.0001
R-squared	=	0.0016

Root MSE = 194.4

	earn4	Coef.	Robust Std. Err.	t	P> t	[95% Conf
assignment						
Randomly assigned to Job Corps		16.05513	4.073534	3.94	0.000	8.070101
_cons		197.9258	3.072095	64.43	0.000	191.9039

We see simple difference in outcomes is about 16 dollars higher, which is what we see in the Schochet, P. Z., Burghardt, J., Glazerman, S. (2001): “National Job Corps study: The impacts of Job Corps on participants’ employment and related outcomes”, Mathematica Policy Research, Washington, DC.

2.1.3 Covariate Balance Test and Specification Tests

Next, let’s test the is there are any covariates associated with the treatment after randomization

```
ttest age, by(assignment)
ttest educ, by(assignment)
ttest educmum, by(assignment)
ttest educdad, by(assignment)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Randomly	3,663	18.33825	.034554	2.091302	18.2705	18.40599
Randomly	5,577	18.50099	.0291468	2.17666	18.44385	18.55813
combined	9,240	18.43647	.0223105	2.144592	18.39274	18.48021
diff		-.1627389	.0455812		-.2520881	-.0733896
diff = mean(Randomly) - mean(Randomly)				t = -3.5703		
Ho: diff = 0				degrees of freedom = 9238		
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 0.0002		Pr(T > t) = 0.0004		Pr(T > t) = 0.9998		

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Randomly	3,663	9.924379	.0327304	1.980934	9.860207	9.988551
Randomly	5,577	9.977228	.0270215	2.017948	9.924255	10.0302
combined	9,240	9.956277	.0208418	2.003416	9.915423	9.997132
diff		-.052849	.0426065		-.1363671	.0306691
diff = mean(Randomly) - mean(Randomly)				t = -1.2404		
Ho: diff = 0				degrees of freedom = 9238		
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 0.1074		Pr(T > t) = 0.2149		Pr(T > t) = 0.8926		

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Randomly	3,663	9.280917	.084987	5.143648	9.114291	9.447544
Randomly	5,577	9.335664	.0675206	5.042389	9.203298	9.468031
combined	9,240	9.313961	.0528746	5.082565	9.210315	9.417607
diff		-.0547471	.108098		-.266643	.1571489
diff = mean(Randomly) - mean(Randomly)				t = -0.5065		
Ho: diff = 0				degrees of freedom = 9238		
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 0.3063		Pr(T > t) = 0.6125		Pr(T > t) = 0.6937		

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Randomly	3,663	7.019383	.1004599	6.080105	6.82242	7.216346
Randomly	5,577	7.04662	.0803195	5.998205	6.889163	7.204078
combined	9,240	7.035823	.062736	6.030492	6.912846	7.158799
diff		-.027237	.1282603		-.2786556	.2241816

```

-----
diff = mean(Randomly) - mean(Randomly)                                t = -0.2124
Ho: diff = 0                                                            degrees of freedom = 9238

Ha: diff < 0                    Ha: diff != 0                    Ha: diff > 0
Pr(T < t) = 0.4159              Pr(|T| > |t|) = 0.8318          Pr(T > t) = 0.5841

```

We fail to reject the null hypothesis that there are differences between treatment and control among the covariates of age, education attained, mother's education, and father's education.

```

tab black assignment, chi2
tab female assignment, chi2
tab hispanic assignment, chi2
tab cohabmarried assignment, chi2
tab haschild assignment, chi2

```

Individual	Random assignment to		
is Black	Job Corps		
or African	Randomly	Randomly	Total
American	Randomly	Randomly	Total
-----+-----+-----			
Non-Black	1,873	2,810	4,683
Black	1,790	2,767	4,557
-----+-----+-----			
Total	3,663	5,577	9,240

Pearson chi2(1) = 0.4941 Pr = 0.482

Individual	Random assignment to		
is female	Job Corps		
Male	Randomly	Randomly	Total
Female	Randomly	Randomly	Total
-----+-----+-----			
Male	2,220	2,960	5,180
Female	1,443	2,617	4,060
-----+-----+-----			
Total	3,663	5,577	9,240

Pearson chi2(1) = 50.9039 Pr = 0.000

Individual is	Random assignment to		
Latino/Hispanic	Job Corps		
Non-Latino	Randomly	Randomly	Total
-----+-----+-----			
Non-Latino	3,024	4,641	7,665

Latino/Hispanic	639	936	1,575
-----+			
Total	3,663	5,577	9,240

Pearson chi2(1) = 0.6842 Pr = 0.408			

Individual is Married	Random assignment to		
or Cohabiting at	Job Corps		
assignment	Randomly	Randomly	Total
-----+			
Not Married or cohabi	3,441	5,233	8,674
Married or Cohabiting	222	344	566
-----+			
Total	3,663	5,577	9,240

Pearson chi2(1) = 0.0445 Pr = 0.833			

Individual has child	Random assignment to		
or children at	Job Corps		
assignment	Randomly	Randomly	Total
-----+			
No children	2,981	4,417	7,398
Has at least one chil	682	1,160	1,842
-----+			
Total	3,663	5,577	9,240

Pearson chi2(1) = 6.5895 Pr = 0.010			

Individual	Random assignment to		
has ever	Job Corps		
worked at	Randomly	Randomly	Total
assignment	-----+		
Never worked	3,155	4,767	7,922
Has worked	508	810	1,318
-----+			
Total	3,663	5,577	9,240

Pearson chi2(1) = 0.7768 Pr = 0.378			

We see that we fail to reject the null hypothesis that there is a difference between treatment and control for individuals for Black, Latino/Hispanic, Married/Cohabiting, and Ever Worked. However, we reject the null hypothesis that

there is no difference between treatment and control for being female or having any children.

Should we be concerned? Yes!

Let's do a sensitivity tests. The first model will be the simple difference, while the second model will include the covariates we rejected the null hypothesis. The third model will include all covariates.

While there is some concern for a lack of covariate balance, the full data set utilizes weights which we do not have in the public use data.

*What happens to the standard errors

```
est clear
eststo m1: quietly reg earny4 i.assignment
eststo m2: quietly reg earny4 i.assignment i.female i.haschild
eststo m3: quietly reg earny4 i.assignment i.female i.black i.hispanic i.cohabmarried
esttab m1 m2 m3, mtitle("Model 1" "Model 2" "Model 3") keep(1.assignment)
```

	(1) Model 1	(2) Model 2	(3) Model 3
1.assignment	16.06*** (3.88)	21.07*** (5.16)	19.59*** (4.96)
N	9240	9240	9240

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

We see that our ATE increase estimates range from 16 dollars to 21 dollars.

2.2 Leaflet Information Evaluation

Next we will look at an impact evaluation of an information leaflet about coffee production on students' awareness. The target population is Bulgarian high school and high school students and the outcome is the awareness of environmental issues due to coffee production.

2.2.1 Inspect the Data

Let's inspect the data by running some descriptive statistics. Our dependent variable is awarewaste, while treatment is a leaflet treatment is randomly assigned to students.

```
cd "/Users/Sam/Desktop/Econ 672/Data"
use CoffeeLeaflet, clear

tab awarewaste
tab treatment
```

```
/Users/Sam/Desktop/Econ 672/Data
```

```
(Written by R.          )
```

Awareness of waste production due to coffee production	Freq.	Percent	Cum.
Not Aware	171	33.93	33.93
Slightly Not Aware	157	31.15	65.08
Neither	121	24.01	89.09
Slightly Aware	41	8.13	97.22
Fully Aware	14	2.78	100.00
Total	504	100.00	

Treatment Status	Freq.	Percent	Cum.
Control Group	261	50.00	50.00
Leaflet Treatment Group	261	50.00	100.00
Total	522	100.00	

2.2.2 Estimate the impact

Let's run the simple difference in outcomes with *ttest* command.

```
ttest awarewaste, by(treatment)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
Control	254	2.007874	.0628675	1.001943	1.884064	2.131684
Leaflet	250	2.288	.0702564	1.110852	2.149627	2.426373
combined	504	2.146825	.0474646	1.065578	2.053572	2.240079
diff		-.280126	.0942007		-.4652021	-.0950498
diff = mean(Control) - mean(Leaflet)				t = -2.9737		
Ho: diff = 0				degrees of freedom = 502		
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 0.0015		Pr(T > t) = 0.0031		Pr(T > t) = 0.9985		

We reject the null hypothesis that the leaflet treatment had no impact, and it increases awareness by 0.28.

Let's run an OLS regression.

```
reg awarewaste i.treatment
```

Source	SS	df	MS	Number of obs	=	504
Model	9.88666867	1	9.88666867	F(1, 502)	=	8.84
Residual	561.248252	502	1.11802441	Prob > F	=	0.0031
Total	571.134921	503	1.1354571	R-squared	=	0.0173
				Adj R-squared	=	0.0154
				Root MSE	=	1.0574

awarewaste	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
treatment						
Leaflet Treatment Group	.280126	.0942007	2.97	0.003	.0950498	.4652021
_cons	2.007874	.0663451	30.26	0.000	1.877526	2.138222

Wait a second! What does 0.28 mean? What is a potential problem here?

Look at the outcome variable again. It is an ordinal variable! Let's use a χ^2 test

```
tab awarewaste treatment, chi2
```

Awareness of waste production due to Treatment Status coffee production Control G Leaflet T Total			
-----+-----+-----+-----			
Not Aware	97	74	171
Slightly Not Aware	82	75	157
Neither	56	65	121
Slightly Aware	14	27	41
Fully Aware	5	9	14
-----+-----+-----+-----			
Total	254	250	504

Pearson chi2(4) = 9.3087 Pr = 0.054

Let's run an ordinal logit regression.

```
ologit awarewaste i.treatment, or
```

```
Iteration 0: log likelihood = -693.62947
Iteration 1: log likelihood = -689.57824
Iteration 2: log likelihood = -689.57505
Iteration 3: log likelihood = -689.57505
```

Ordered logistic regression	Number of obs	=	504
	LR chi2(1)	=	8.11
	Prob > chi2	=	0.0044
Log likelihood = -689.57505	Pseudo R2	=	0.0058

awarewaste	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
treatment						
Leaflet Treatment Group	.4609564	.1623425	2.84	0.005	.1427709	.7791419
-----+-----						
/cut1	-.4493223	.1211244			-.6867218	-.2119229
/cut2	.8543491	.1252889			.6087873	1.099911
/cut3	2.346133	.168949			2.014999	2.677267
/cut4	3.80845	.2865533			3.246816	4.370084
-----+-----						

Individuals who received treatment were 1.46 times more likely to be aware of the effect of coffee production on environmental issues than those in the control group.

2.2.3 Covariate Balance Test

Let's test covariate balance

```
foreach v of varlist sex mumedu dadedu bulgnationality drinkcoffee citysofia {
  tab awarewaste `v', chi2
}
```

Awareness of waste	sex		
production due to			
coffee production	0	1	Total
Not Aware	94	77	171
Slightly Not Aware	102	53	155
Neither	65	55	120
Slightly Aware	22	19	41
Fully Aware	3	11	14
Total	286	215	501

Pearson chi2(4) = 13.0039 Pr = 0.011

Awareness of waste	mumedu			
production due to				
coffee production	1	2	3	Total
Not Aware	3	69	88	160
Slightly Not Aware	5	48	98	151
Neither	1	31	76	108
Slightly Aware	1	9	27	37
Fully Aware	0	3	9	12
Total	10	160	298	468

Pearson chi2(8) = 11.8017 Pr = 0.160

Awareness of waste	dadedu			
production due to				
coffee production	1	2	3	Total
Not Aware	4	67	83	154
Slightly Not Aware	2	53	87	142
Neither	0	42	62	104
Slightly Aware	0	9	30	39
Fully Aware	0	3	8	11

Total	6	174	270	450
-------	---	-----	-----	-----

Pearson $\chi^2(8) = 10.9594$ Pr = 0.204

Awareness of waste			
production due to	bulgnationality		
coffee production	0	1	Total
Not Aware	4	167	171
Slightly Not Aware	5	152	157
Neither	2	119	121
Slightly Aware	0	41	41
Fully Aware	0	14	14
Total	11	493	504

Pearson $\chi^2(4) = 2.1444$ Pr = 0.709

Awareness of waste						
production due to		drinkcoffee				
coffee production	1	2	3	4	5	Total
Not Aware	68	18	21	32	31	170
Slightly Not Aware	39	25	32	27	34	157
Neither	46	19	19	21	14	119
Slightly Aware	12	5	4	12	8	41
Fully Aware	4	0	4	3	2	13
Total	169	67	80	95	89	500

Pearson $\chi^2(16) = 23.6155$ Pr = 0.098

Awareness of waste			
production due to	citysofia		
coffee production	0	1	Total
Not Aware	46	125	171
Slightly Not Aware	40	117	157
Neither	27	94	121
Slightly Aware	7	34	41
Fully Aware	2	12	14
Total	122	382	504

Pearson $\chi^2(4) = 2.9391$ Pr = 0.568

From our covariate balance test, we have a slight concern for imbalance in sex. However, all other characteristics were not correlated with treatment status.

```
est clear
quietly {
eststo m1: ologit awarewaste i.treatment, or
eststo m2: ologit awarewaste i.treatment i.mumedu i.sex, or
eststo m3: ologit awarewaste i.treatment i.drinkcoffee i.sex i.mumedu i.dadedu i.bulgn
}

esttab m1 m2 m3, mtitle("Model 1" "Model 2" "Model 3") keep(1.treatment)
```

	(1) Model 1	(2) Model 2	(3) Model 3
awarewaste			
1.treatment	0.461** (2.84)	0.566*** (3.32)	0.583** (3.27)
N	504	466	434

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Our estimates range from an odd ratio of 1.46 to 1.58. Notice that as we add covariates, our sample size is dropping, which means not all observations provided characteristic information.

Chapter 3

Randomized Inference

Adapted from Cunningham (2021)

We will use a placebo-based test to calculate exact p-values. We need every single permutation of treatment to calculate the exact p-values from the ATE.

3.1 Set up the Dataset

First we will set the seed so we can replicate our randomized results

```
set seed 1234
```

Next we will bring in our

```
cd "/Users/Sam/Desktop/Econ 672/Data"  
use ri.dta, clear  
tempfile ri  
gen id = _n  
save "`ri'", replace
```

name[1]		Andy					
	name	d	y	y0	y1	id	
1	Andy	1	10	.	10	1	
2	Ben	1	5	.	5	2	
3	Chad	1	16	.	16	3	
4	Daniel	1	3	.	3	4	
5	Edith	0	5	5	.	5	
6	Frank	0	7	7	.	6	
7	George	0	8	8	.	7	
8	Hank	0	10	10	.	8	

Next we will create all combinations from our eight observations. We will need to install the *percom* package into stata.

```
ssc install percom
```

Out of 8 observations, we get 70 permutations. We can use the *combin* command to get these permutations. From the *help combin*, the *combin* command generates k unique combinations from a set of n objects, where k is not greater than 5 ($2 \leq k \leq 5$). Program first arranges k possible permutations and then retains unique combinations. It supports both string and numeric variables.

combination: $\frac{n!}{k!(n-k)!}$

We will tell the *combin* command that we want 4 combinations from our 8 observations.

```
combin id, k(4)
```

```

      k =          4
      N =          8
Combinations Formed: 70

```

id_2[18]		3				
	id_1	id_2	id_3	id_4		
1	1	2	3	4		
2	1	2	3	5		
3	1	2	3	6		
4	1	2	3	7		
5	1	2	3	8		
6	1	2	4	5		
7	1	2	4	6		
8	1	2	4	7		
9	1	2	4	8		
10	1	2	5	6		
11	1	2	5	7		
12	1	2	5	8		
13	1	2	6	7		
14	1	2	6	8		
15	1	2	7	8		
16	1	3	4	5		
17	1	3	4	6		
18	1	3	4	7		
19	1	3	4	8		
20	1	3	5	6		
21	1	3	5	7		
22	1	3	5	8		
23	1	3	6	7		
24	1	3	6	8		
25	1	3	7	8		
26	1	4	5	6		
27	1	4	5	7		
28	1	4	5	8		
29	1	4	6	7		
30	1	4	6	8		
31	1	4	7	8		

Next we will label the permutations 1 to 70.

```

gen permutation = _n
tempfile combo
save "`combo'", replace

forvalue i =1/4 {
    rename id_`i' treated`i'
}

```

Next we will merge our permutations with our dataset of potential outcomes

```

destring treated*, replace
cross using `ri'

```

	treated1	treated2	treated3	treated4	permutation	name	d	y	y0	y1	id
1	1	2	3	4	1	Andy	1	10	.	10	
2	1	2	3	4	1	Ben	1	5	.	5	
3	1	2	3	4	1	Chad	1	16	.	16	
4	1	2	3	4	1	Daniel	1	3	.	3	
5	1	2	3	4	1	Edith	0	5	.	5	
6	1	2	3	4	1	Frank	0	7	.	7	
7	1	2	3	4	1	George	0	8	.	8	
8	1	2	3	4	1	Hank	0	10	.	10	
9	1	2	3	5	2	Andy	1	10	.	10	
10	1	2	3	5	2	Ben	1	5	.	5	
11	1	2	3	5	2	Chad	1	16	.	16	
12	1	2	3	5	2	Daniel	1	3	.	3	
13	1	2	3	5	2	Edith	0	5	.	5	
14	1	2	3	5	2	Frank	0	7	.	7	
15	1	2	3	5	2	George	0	8	.	8	
16	1	2	3	5	2	Hank	0	10	.	10	
17	1	2	3	6	3	Andy	1	10	.	10	
18	1	2	3	6	3	Ben	1	5	.	5	
19	1	2	3	6	3	Chad	1	16	.	16	
20	1	2	3	6	3	Daniel	1	3	.	3	
21	1	2	3	6	3	Edith	0	5	.	5	
22	1	2	3	6	3	Frank	0	7	.	7	
23	1	2	3	6	3	George	0	8	.	8	
24	1	2	3	6	3	Hank	0	10	.	10	
25	1	2	3	7	4	Andy	1	10	.	10	
26	1	2	3	7	4	Ben	1	5	.	5	
27	1	2	3	7	4	Chad	1	16	.	16	
28	1	2	3	7	4	Daniel	1	3	.	3	
29	1	2	3	7	4	Edith	0	5	.	5	
30	1	2	3	7	4	Frank	0	7	.	7	
31	1	2	3	7	4	George	0	8	.	8	

Figure 3.1: Merge Dataset with Permutations

3.2 Randomize Assignment to Treatment

Next we will randomized treatment and calculate other test statistics.

First, we create a fake treatment variable $d2$

```

gen d2 = .
replace d2 = 1 if id == treated1 | id == treated2 | id == treated3 | id == treated4
replace d2 = 0 if ~(id == treated1 | id == treated2 | id == treated3 | id == treated4)
gen check = d - d2

```

treated1[14] 1									
	name	d	y	y0	y1	id	d2	check	
1	Andy	1	10	.	10	1	1	0	
2	Ben	1	5	.	5	2	1	0	
3	Chad	1	16	.	16	3	1	0	
4	Daniel	1	3	.	3	4	1	0	
5	Edith	0	5	5	.	5	0	0	
6	Frank	0	7	7	.	6	0	0	
7	George	0	8	8	.	7	0	0	
8	Hank	0	10	10	.	8	0	0	
9	Andy	1	10	.	10	1	1	0	
10	Ben	1	5	.	5	2	1	0	
11	Chad	1	16	.	16	3	1	0	
12	Daniel	1	3	.	3	4	0	1	
13	Edith	0	5	5	.	5	1	-1	
14	Frank	0	7	7	.	6	0	0	
15	George	0	8	8	.	7	0	0	
16	Hank	0	10	10	.	8	0	0	
17	Andy	1	10	.	10	1	1	0	
18	Ben	1	5	.	5	2	1	0	
19	Chad	1	16	.	16	3	1	0	
20	Daniel	1	3	.	3	4	0	1	
21	Edith	0	5	5	.	5	0	0	
22	Frank	0	7	7	.	6	1	-1	

Figure 3.2: Set up False Treatment

Next, we will calculate true effect using absolute value of simple difference in outcomes (*SDO*).

```
egen te1 = mean(y) if d2==1, by(permutation)
egen te0 = mean(y) if d2==0, by(permutation)
```

treated1[14] 1									
	name	d	y	y0	y1	id	d2	check	
1	Andy	1	10	.	10	1	1	0	
2	Ben	1	5	.	5	2	1	0	
3	Chad	1	16	.	16	3	1	0	
4	Daniel	1	3	.	3	4	1	0	
5	Edith	0	5	5	.	5	0	0	
6	Frank	0	7	7	.	6	0	0	
7	George	0	8	8	.	7	0	0	
8	Hank	0	10	10	.	8	0	0	
9	Andy	1	10	.	10	1	1	0	
10	Ben	1	5	.	5	2	1	0	
11	Chad	1	16	.	16	3	1	0	
12	Daniel	1	3	.	3	4	0	1	
13	Edith	0	5	5	.	5	1	-1	
14	Frank	0	7	7	.	6	0	0	
15	George	0	8	8	.	7	0	0	
16	Hank	0	10	10	.	8	0	0	
17	Andy	1	10	.	10	1	1	0	
18	Ben	1	5	.	5	2	1	0	
19	Chad	1	16	.	16	3	1	0	
20	Daniel	1	3	.	3	4	0	1	
21	Edith	0	5	5	.	5	0	0	
22	Frank	0	7	7	.	6	1	-1	
23	George	0	8	8	.	7	0	0	
24	Hank	0	10	10	.	8	0	0	

Figure 3.3: Get the average treatment effects by permutations

3.3 Calculate the exact p-value

Get average treatment effects (*ATE*)

```
collapse (mean) te1 te0, by(permutation)
gen      ate = te1 - te0
keep     ate permutation
```

Next, we rank the permutations

```
gsort -ate
gen rank = _n
sum rank if permutation==1
```

Finally, we calculate Exact p -value

```
gen pvalue = (`r(mean)'/70)
list pvalue if permutation==1
```

```
      |      pvalue |
      |-----|
29.  | .41428571 |
      +-----+
```

Cunningham, S. (2021). *Causal Inference: The Mixtape Tape*. Yale University Press

Estout. (2025). *Making Regression Tables in Stata*. Retrieved from: <https://repec.sowi.unibe.ch/stata/estout/>

Huber, M. (2023). *Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R*. The MIT Press.

Schochet, P. Z., Burghardt, J., Glazerman, S. (2001): “National Job Corps study: The impacts of Job Corps on participants’ employment and related outcomes”, Mathematica Policy Research, Washington, DC.

permutation[14]		14		
	permutation	ate		
1	1	1		
2	2	2		
3	3	3		
4	4	3.5		
5	5	4.5		
6	6	-4.5		
7	7	-3.5		
8	8	-3		
9	9	-2		
10	10	-2.5		
11	11	-2		
12	12	-1		
13	13	-1		
14	14	0		
15	15	.5		
16	16	1		
17	17	2		

Figure 3.4: Collapse the treatment effects by permutation

permutation[14]		3		
	permutation	ate	rank	
1	25	6	1	
2	24	5.5	2	
3	65	4.5	3	
4	22	4.5	4	
5	23	4.5	5	
6	5	4.5	6	
7	21	3.5	7	
8	64	3.5	8	
9	4	3.5	9	
10	45	3.5	10	
11	19	3.5	11	
12	63	3	12	
13	44	3	13	
14	3	3	14	
15	20	3	15	

Figure 3.5: Sort and Rank by ATE